Spectral Radius Minimization for Signed Squares of Cycles

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Abstract

For even $n \geq 8$, let $C_n^2 = C_n(1,2)$ be the square of the cycle, with its distance-one and distance-two edges independently assigned signs, with no periodicity assumption on the signing, and let m_n be the minimum signed adjacency spectral radius. A distinguished twisted signing attains $\rho_-(n)$, where $\rho_-(n)^2 = 4 + 2\cos(2\pi/n) + 2\cos(4\pi/n)$, at every even order. We prove that equality holds exactly for $n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}$; equivalently, strict failure occurs precisely at 32, at 40, and at every even $n \geq 48$. Structurally, a period-eight reference phase has squared spectral edge $\eta < 8$, while its six-gap phase slip is the unique least-edge abnormal positive single gap and has an isolated rank-two squared edge $c_6 \in (\eta, 8)$. Charge-compatible separated copies of this slip, together with a discrete IMS estimate, prove eventual failure, while exact finite verification shows that continuous failure begins precisely at order 48. The finite portions are reduced to explicitly finite objects and independently verified using exact arithmetic, so floating-point evidence makes no spectral decision.

Keywords: signed graph; spectral radius; cycle square; switching equivalence; phase slip; computer-assisted proof

MSC 2020: 05C50 (primary); 05C22, 68V05 (secondary)

1 Introduction

Let $n \geq 8$ be even. We write

$$G_n = C_n^2 = C_n(1,2)$$

for the graph on $\mathbb{Z}/n\mathbb{Z}$ in which two vertices are adjacent when their cyclic distance is one or two. An edge signing is a map $\sigma : E(G_n) \to \{\pm 1\}$, and A_σ is the real symmetric matrix whose i, j entry is $\sigma(ij)$ on an edge and zero otherwise. Our problem is to determine

$$m_n = \min_{\sigma : E(G_n) \to \{\pm 1\}} \rho(A_\sigma).$$

Thus the already formed cycle square is being signed: the distance-one and distance-two edges receive signs independently. It is not the graph obtained by first signing a cycle and then squaring that signed object. Switching at vertices conjugates A_σ by a diagonal $\{\pm 1\}$-matrix, so the minimization naturally takes place over switching classes.

The problem belongs to the general signing program in spectral graph theory. Signings encode the new eigenvalues of two-lifts in the work of Bilu and Linial [3], while Belardo, Cioabă, Koolen, and Wang ask explicitly which signatures minimize the adjacency spectral radius of a fixed connected graph [2]. Interlacing families give strong existence results for one side of the signed spectrum; in the bipartite setting spectral symmetry turns this into the two-sided control used to construct Ramanujan lifts [9]. The graphs C_n^2 are nonbipartite, however, so a one-sided eigenvalue estimate does not by itself control $\rho(A_\sigma)$. Here the graph is fixed at each order and the question is the exact two-sided optimum, rather than an existence bound uniform over a broad graph class.

The problem-specific starting point is Suvagiya's preprint [10, Conjecture 3]. For $C_n(1,2)$, it identifies four distinguished switching classes in which every quadrilateral is unbalanced, computes the spectra of the two twisted classes, verifies global optimality for even orders through 18, and formulates the all-even-order assertion as Conjecture 3. We determine exactly when that twisted candidate is globally minimizing, thereby resolving and disproving the conjecture. In particular, the problem, the flux coordinates, and the twisted spectral formula are inputs from that work, not novelty claims of the present paper.

The answer is unexpectedly nonmonotone. The proposed equality holds for every even order from 8 through 30, fails first at 32, returns at $34, 36, 38$, fails again at 40, and returns once more at $42, 44, 46$. Only from 48 onward does failure persist at every even order. The first counterexample therefore does not mark the beginning of continuous failure, and the two isolated failures cannot be inferred from the eventual phase-slip mechanism developed below.

To state the classification, set

$$\rho_-(n)^2 = 4 + 2\cos\frac{2\pi}{n} + 2\cos\frac{4\pi}{n}, \qquad \theta_n = \rho_-(n)^2.$$

The distinguished class has alternating triangle flux and negative Hamilton-cycle holonomy; a direct antiperiodic Fourier calculation shows that it attains $\rho_-(n)$ for every even $n \geq 8$. This gives the upper bound $m_n \leq \rho_-(n)$ at all orders. At an order where equality holds, the reverse inequality for every signing is a separate conclusion of exact exhaustion.

Theorem 1.1 (Complete classification)**.** *For every even $n \geq 8$, the distinguished twisted signing attains $\rho_-(n)$, and*

$$m_n < \rho_-(n) \quad \Longleftrightarrow \quad n = 32, \quad n = 40, \quad \text{or} \quad n \geq 48.$$

Equivalently,

$$m_n = \rho_-(n) \quad \Longleftrightarrow \quad n \in \{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 36, 38, 42, 44, 46\}.$$

Theorem 1.1 classifies the truth of the proposed equality. It neither classifies all minimizing signings at the valid orders nor determines the exact value of m_n at the failing orders.

Its finite component determines the irregular small-order truth table. A different part of the argument explains why failure is eventually uninterrupted: a low-spectral periodic phase supports localized elementary interfaces, and well-separated copies of those interfaces can be closed on large rings. The assertion that continuous failure begins at 48 uses both this analytic tail and an exact finite bridge; the interface theory alone gives no such sharp onset.

The structural comparison phase is period eight and is distinct from the twisted signing that attains $\rho_-(n)$. In its quadrilateral-flux word, the positive sites occur at spacing four. Replacing one spacing by a positive gap g creates a single interface; we call $q = g - 4$ its excess charge. For a cyclic collection of gaps, the global closing law is $\sum_j q_j \equiv n \pmod 8$, whereas the local translation sector carried by one charge is $\sigma_{\text{sec}}(q) = q \pmod 4$. These congruences have different roles and are not interchangeable. In particular, one, two, and three copies of the gap-six interface, whose charge is two, supply the three nonzero even residue classes needed in the finite-ring constructions.

The reference squared spectral edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}.$$

For the bilateral positive single-gap operator A_g, put $H_g = A_g^2$ and $E(g) = \sup \operatorname{Spec}(H_g)$. Define c_6 to be the unique root in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}$$

of the degree-ten polynomial

$$\begin{aligned} p_6(y) = 16y^{10} &- 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 \\ &- 423904y^5 + 484528y^4 - 270464y^3 + 137856y^2 \\ &- 19968y + 256. \end{aligned}$$

The exact isolation is important: all later strict comparisons are made against rational endpoints, not against a decimal approximation.

Theorem 1.2 (Reference and single-gap spectral hierarchy)**.** *For both lifts and both orientations,*

$$E(4) = \eta < c_6 = E(6), \qquad \mathrm{Spec}_{\mathrm{ess}}(H_6) = \mathrm{Spec}(H_{\mathrm{ref}}), \qquad \dim\ker(H_6 - c_6) = 2.$$

Moreover, for every positive integer $g \notin \{4, 6\}$,

$$E(g) > c_6 + \frac{1}{250}.$$

The scope of Theorem 1.2 is precise: G6 is the unique minimizer among abnormal positive single gaps. It is not asserted to minimize over arbitrary multi-gap or finite-core interfaces. The strict single-gap hierarchy makes G6 the elementary low-cost defect above the period-eight bulk. Its charge allows one, two, or three separated copies to close in the required residue classes. Exact patch identification transfers the bilateral reference/G6 bounds to those finite rings, and a discrete IMS partition controls the squared spectral top between the patches. This proves failure for every even $n \geq 240$. Exact finite verification on $48 \leq n < 240$ then shows that continuous failure begins precisely at 48.

The computer-assisted part is organized as a proof by finite reduction, not as numerical evidence. Switching quotients and local interlacing reduce the relevant cases to finite exact objects. For $8 \leq n \leq 32$, complete switching-class decisions establish the lower bounds through 30 and an exact witness at 32. For $34, 36, 38, 42, 44, 46$, a parity-lifted finite state graph is proved sound, complete, and terminating, and every terminal case is closed exactly; a separate rational positive-definiteness certificate gives the witness at 40. Finally, rational positive-definiteness certificates cover every even order from 48 through 238. Independent checkers reconstruct the finite objects and repeat the integer, rational, or algebraic decisions. Exhaustive generation and local finite-state reductions have established precedents in graph theory [7, 5], while all-order classifications often combine an analytic tail with finite closure [8, 6].

Exact signed spectral-radius classifications are known in settings where the underlying graph is also allowed to vary, including unbalanced unicyclic and bicyclic graphs [1] and connected unbalanced signed graphs of fixed order [4]. Those results make it essential to keep the present novelty claim narrow. To the best of our knowledge, no previous work gives an all-order classification of the minimizing behavior over all real edge signings of the fixed circulant family $C_n(1, 2)$. The closest direct source remains Suvagiya's conjecture; the complete truth set and the elementary single-gap mechanism are the two contributions supplied here. The paper is arranged as follows. Section 2 sets up switching coordinates, the twisted attaining candidate, and the period-eight reference phase. Section 3 develops gaps, charges, and the two closing laws. Sections 4 and 5 prove the G6 edge theorem and the complete single-gap hierarchy. Section 6 passes from bilateral interfaces to finite rings by patch identification and discrete IMS localization. Section 7 closes the remaining orders by exact finite arguments, and Section 8 records the resulting perspective. The appendices collect the algebraic G6 certificate and the exact finite-classification boundary.

2 Switching Coordinates and the Reference Phase

2.1 Edge signings, switching, and Hamilton gauge

2.2 Flux words, holonomy, and the attaining candidate

2.3 The squared local operator

2.4 The period-eight reference phase

3 Gaps, Charges, and Translation Sectors

3.1 Cyclic gaps and excess charge

3.2 Translation sectors

3.3 Composition and legal cyclic closure

4 The Elementary Six-Gap Phase Slip

4.1 The bilateral interface and its essential spectrum

4.2 Algebraic candidate and physical matching

4.3 The global G6 edge

4.4 Rank-two symmetry

5 Optimality Among Single-Gap Interfaces

5.1 Canonical single-gap operators

5.2 Exact finite-gap comparisons

5.3 Uniform control of the large gaps

5.4 Reference and single-gap hierarchy

6 Phase Slips on Finite Rings

6.1 Legal residue constructions

6.2 Identification of local patches

6.3 Discrete IMS localization

6.4 Residue upper bounds and the analytic tail

6.5 Structural refinement for separated interfaces

7 Finite Completion of the Classification

7.1 Exact-computation protocol

7.2 Orders 8 through 32

7.3 Orders 34 through 46

7.4 The exact bridge from 48 to 238

7.5 Exhaustive synthesis

8 Concluding Remarks

Data and Code Availability

A Exact Spectral Certification of the Reference Phase and G6

B Exact Finite Classification and Completeness

References

[1] Francesco Belardo, Maurizio Brunetti, and Adriana Ciampella. Unbalanced unicyclic and bicyclic graphs with extremal spectral radius. *Czechoslovak Mathematical Journal*, 71(2):417–433, 2021.

[2] Francesco Belardo, Sebastian M. Cioabă, Jack H. Koolen, and Jianfeng Wang. Open problems in the spectral theory of signed graphs. *The Art of Discrete and Applied Mathematics*, 1(2):P2.10, 2018.

[3] Yonatan Bilu and Nathan Linial. Lifts, discrepancy and nearly optimal spectral gap. *Combinatorica*, 26(5):495–519, 2006.

[4] Maurizio Brunetti and Zoran Stanić. Unbalanced signed graphs with extremal spectral radius or index. *Computational and Applied Mathematics*, 41(3):118, 2022.

[5] Matt DeVos and Robert Šámal. High-girth cubic graphs are homomorphic to the Clebsch graph. *Journal of Graph Theory*, 66(3):241–259, 2011.

[6] Jan Goedgebeur, Jarne Renders, Gábor Wiener, and Carol T. Zamfirescu. K_2-Hamiltonian graphs: II. *Journal of Graph Theory*, 105(4):580–611, 2024.

[7] Jan Goedgebeur and Oliver Schaudt. Exhaustive generation of k-critical H-free graphs. *Journal of Graph Theory*, 87(2):188–207, 2018.

[8] Huiqiu Lin and Bo Ning. A complete solution to the Cvetković–Rowlinson conjecture. *Journal of Graph Theory*, 97(3):441–450, 2021.

[9] Adam W. Marcus, Daniel A. Spielman, and Nikhil Srivastava. Interlacing families I: Bipartite Ramanujan graphs of all degrees. *Annals of Mathematics*, 182(1):307–325, 2015.

[10] Vaibhav Suvagiya. Signed circulants at the Ramanujan bound. arXiv preprint arXiv:2607.18334v1, 2026.